



NVIDIA Transformer Engine

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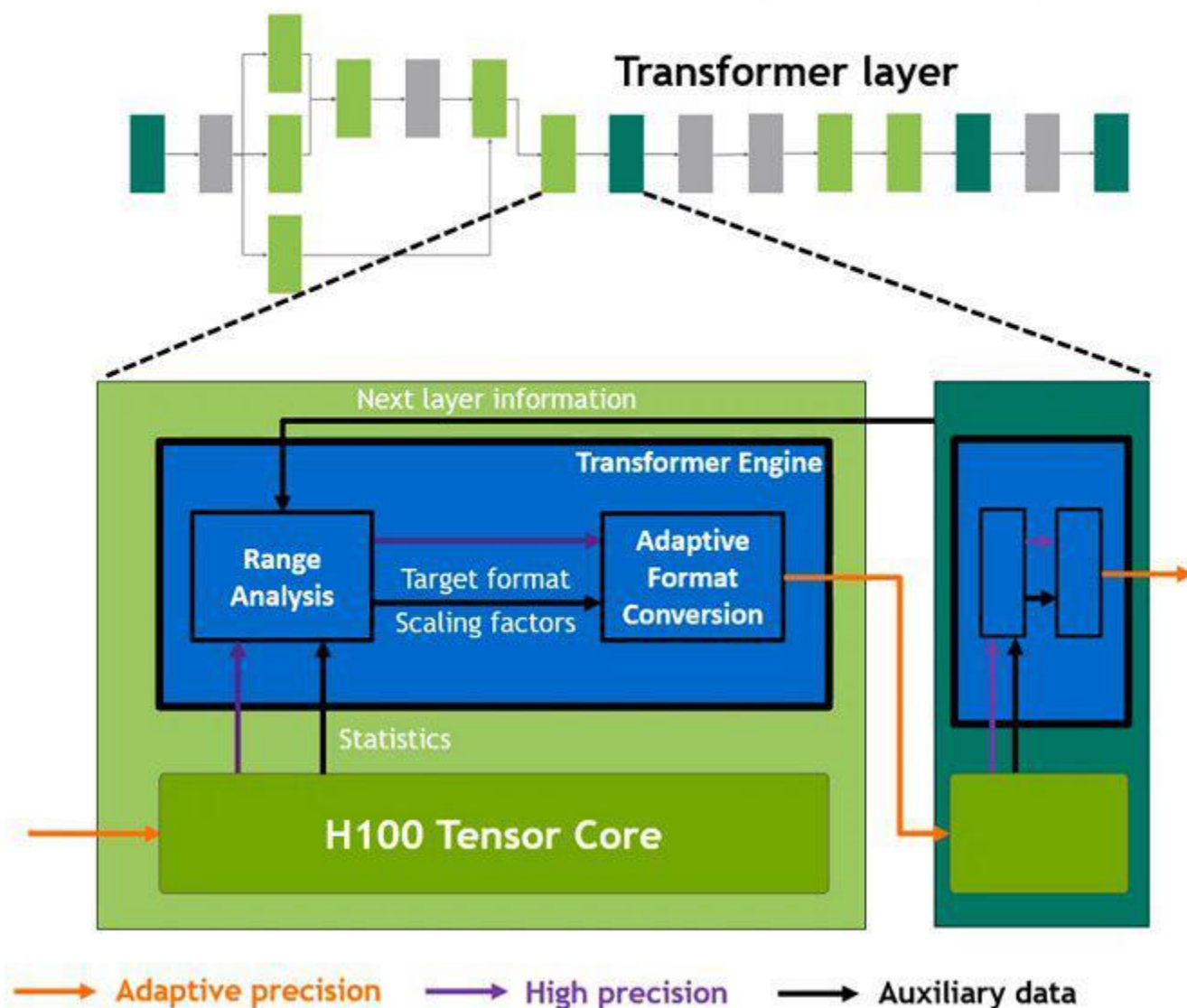
Transformer Engine

Purpose: A library specifically designed to optimize and accelerate Transformer neural networks on NVIDIA GPUs.

It uses techniques like FP8 (8-bit floating point) precision, which can provide a 6x performance boost over FP16 for trillion-parameter models, while maintaining accuracy by intelligently casting data to the appropriate precision.

Custom heuristics for intelligent precision selection between FP8 and FP16, with automatic re-casting and scaling across neural network layers.

TRANSFORMER ENGINE



Optimal Transformer acceleration with Hopper Tensor Core

- **Transparent** to DL frameworks
- User can **enable/disable**
- Selectively **applies new FP8 format** for highest throughput
- **Monitors tensor statistics** and **dynamically adjusts range** to maintain accuracy

Transformer Engine

With Hopper GPU architecture FP8 precision was introduced, which offers improved performance over FP16 with no degradation in accuracy. Although all major deep learning frameworks support FP16, FP8 support is not available natively in frameworks today.

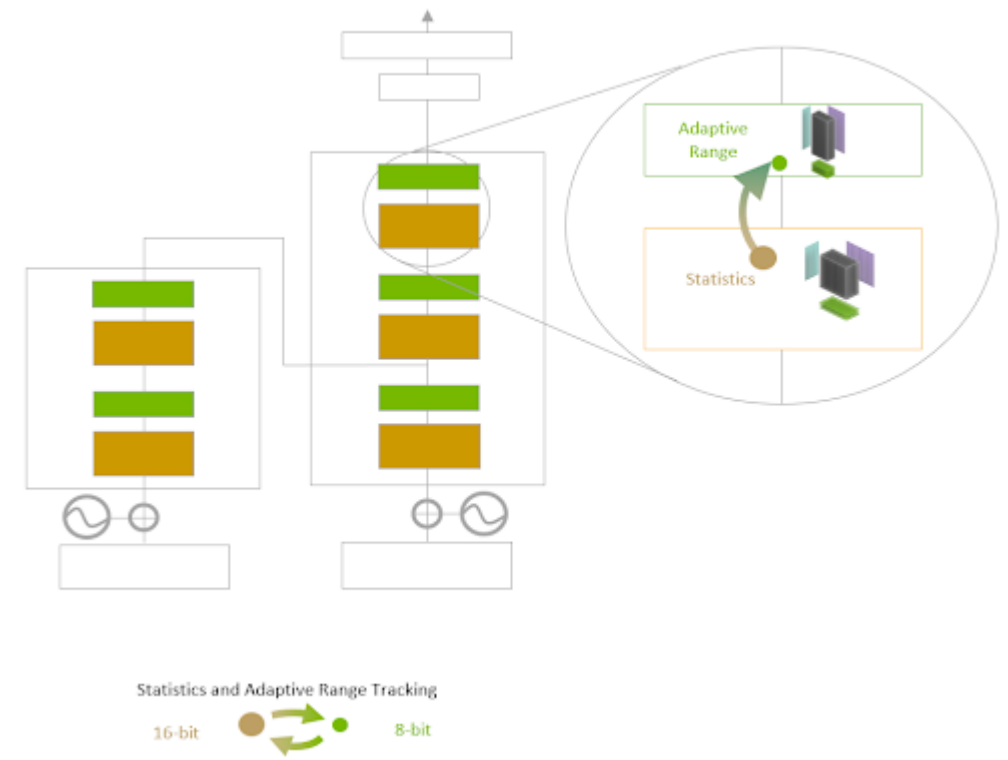
TE addresses the problem of FP8 support by providing APIs that integrate with popular Large Language Model (LLM) libraries.

It provides a Python API consisting of modules to easily build a Transformer layer as well as a framework-agnostic library in C++ including structs and kernels needed for FP8 support.

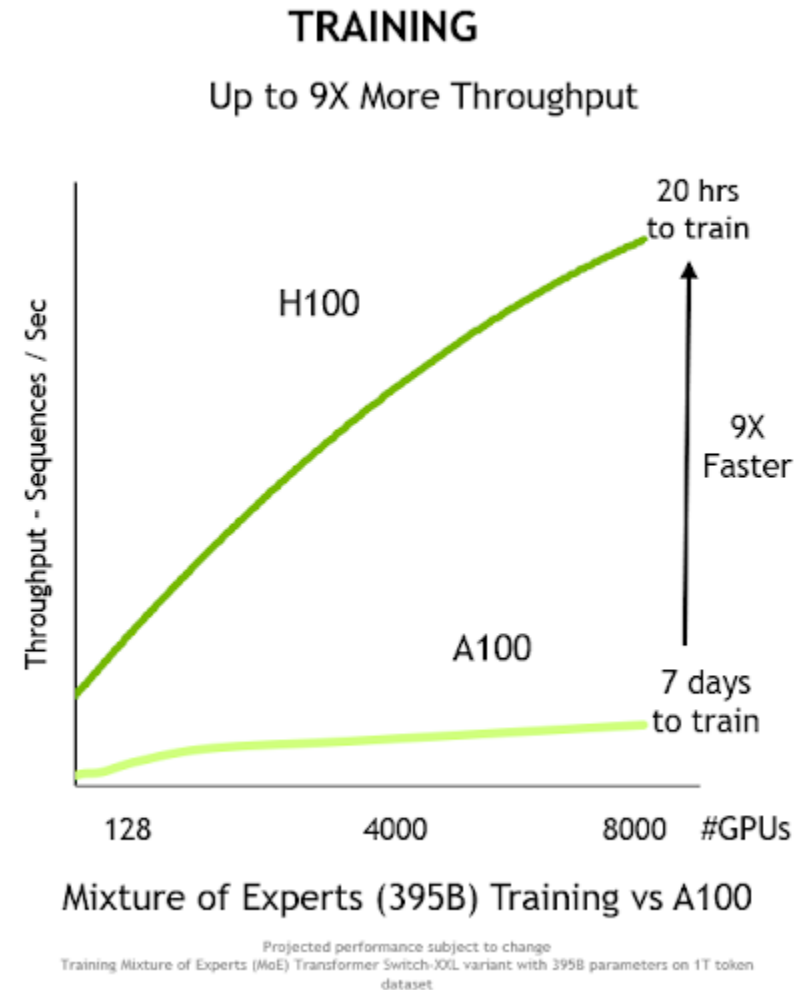
Modules provided by TE internally maintain scaling factors and other values needed for FP8 training, greatly simplifying mixed precision training for users.

Transformer Engine uses per-layer statistical analysis to determine the optimal precision (FP16 or FP8) for each layer of a model, achieving the best performance while preserving model accuracy.

The NVIDIA Hopper architecture advances fourth-generation Tensor Cores by tripling the floating-point operations per second compared with prior-generation TF32, FP64, FP16 and INT8 precisions.



NVIDIA H100 Tensor Core GPU delivers up to 9x more training throughput compared to previous generation, making it possible to train large models in reasonable amounts of time.



Benefits

Higher performance: Dramatically speeds up the training and inference of large language models.

Lower memory: Reduces memory footprint by using lower-precision formats.

Wider model support: Enables the support for larger and more complex models that would otherwise be impractical due to memory or performance constraints.

The Blackwell Transformer Engine utilizes advanced dynamic range management algorithms and fine-grain scaling techniques, called micro-tensor scaling, to optimize inference performance, accuracy, and enable FP4 AI.

This doubles the performance of Blackwell's FP4 Tensor Core, doubles the parameter bandwidth to the HBM memory, and doubles the size of models supported per GPU.

References

<https://github.com/NVIDIA/TransformerEngine>

<https://blogs.nvidia.com/blog/h100-transformer-engine/>